WOODEN DICE RANDOMIZER AND TRAY

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This two-part wooden contraption makes rolling dice more fun. Dice go in through the top of the randomizer — a small enclosed chute lined with brass rods that tumble the dice as they fall — and land in the tray below. The randomizer sits flat inside the tray for easy storage and transport.

The dice randomizer and tray are sized to fit standard polyhedral dice sets and work great for any dice-based game like Yahtzee and D&D. The build plans assume basic woodworking knowledge and access to a table saw, router table, drill press, bandsaw, and spindle sander

WHAT YOU'LL NEED

Supplies

FOR THE TRAY

Tools

- ¼" Router bit
- ⅞" Forstner bit
- $\frac{7}{16}$" Drill bit
- ⅜" Drill bit
- Band clamp
- Spline jig
- Thin-rip guide (opt)
- Flush cut saw

Materials

- Masking tape
- Double-sided tape
- Wood glue
- Gorilla Glue spray
- Felt

FOR THE RANDOMIZER

Tools

- *B*olt cutters

Materials

- ⅛"-dia brass rods
- CA glue & activator
- E6000 adhesive

SANDING & FINISHING

Materials

- 120-grit sandpaper
- 220-grit sandpaper
- Heritage oil

Cut list

Mill boards to ¼" thick.

> ***TIP:*** *resaw ¾" stock on the bandsaw, then run through the planer to get to the final thickness.*

TRAY PARTS. Cut the side blanks oversize so the miters can be trimmed to the final length later:

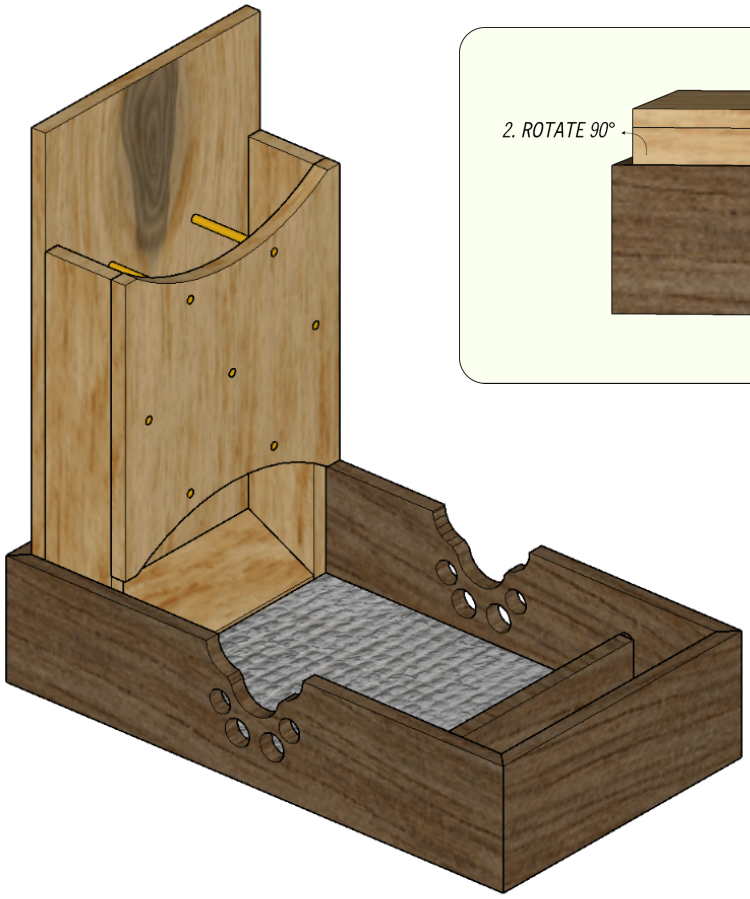
- 2 long sides — 9 ½" x 2" (final length will be 9")
- 2 short sides — 5 ½" x 2" (final length will be 5")
- 1 bottom — 9 ½" x 5 ½" (oversized; final dimensions are cut to fit)
- 1 divider (optional) — 4 ¾" x 1 $\frac{1}{16}$"
- 1 wedge blank — *using ¾" thick stock:* 4 ½" x 3" (oversized; final dimensions are cut to fit)

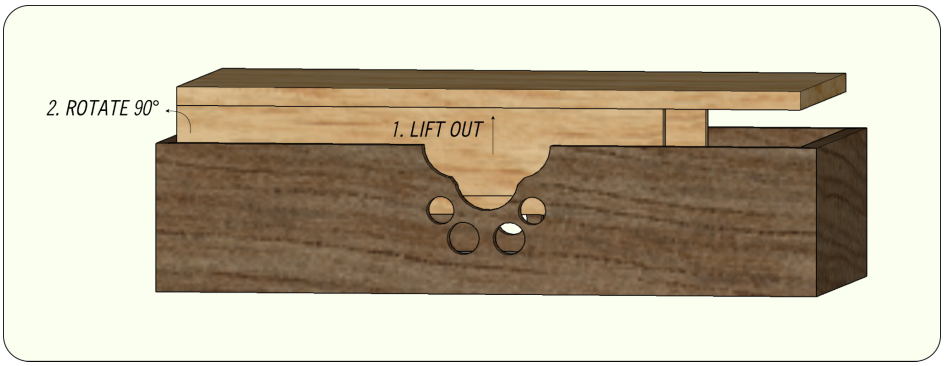
RANDOMIZER PARTS. The randomizer back becomes the visible top lid of the box when the randomizer is stored inside the tray. (Optional: Use a contrasting wood for the randomizer back. You can make the splines on the mitered corners of the tray out of that same contrasting wood for a cohesive look.) Cut randomizer parts to their final sizes:

- 2 sides — 6 $\frac{5}{8}$" x 1 $\frac{3}{16}$"
- 1 back — 8 ½" x 4 ½"
- 1 front — 4 $\frac{11}{16}$" x 4 ½"

Build Sequence

The project is built in five stages: (1) machine and assemble the tray, (2) build the randomizer, including the brass rods, (3) make and install the wedge, (4) complete exterior sanding and finishing, and (5) install the felt.



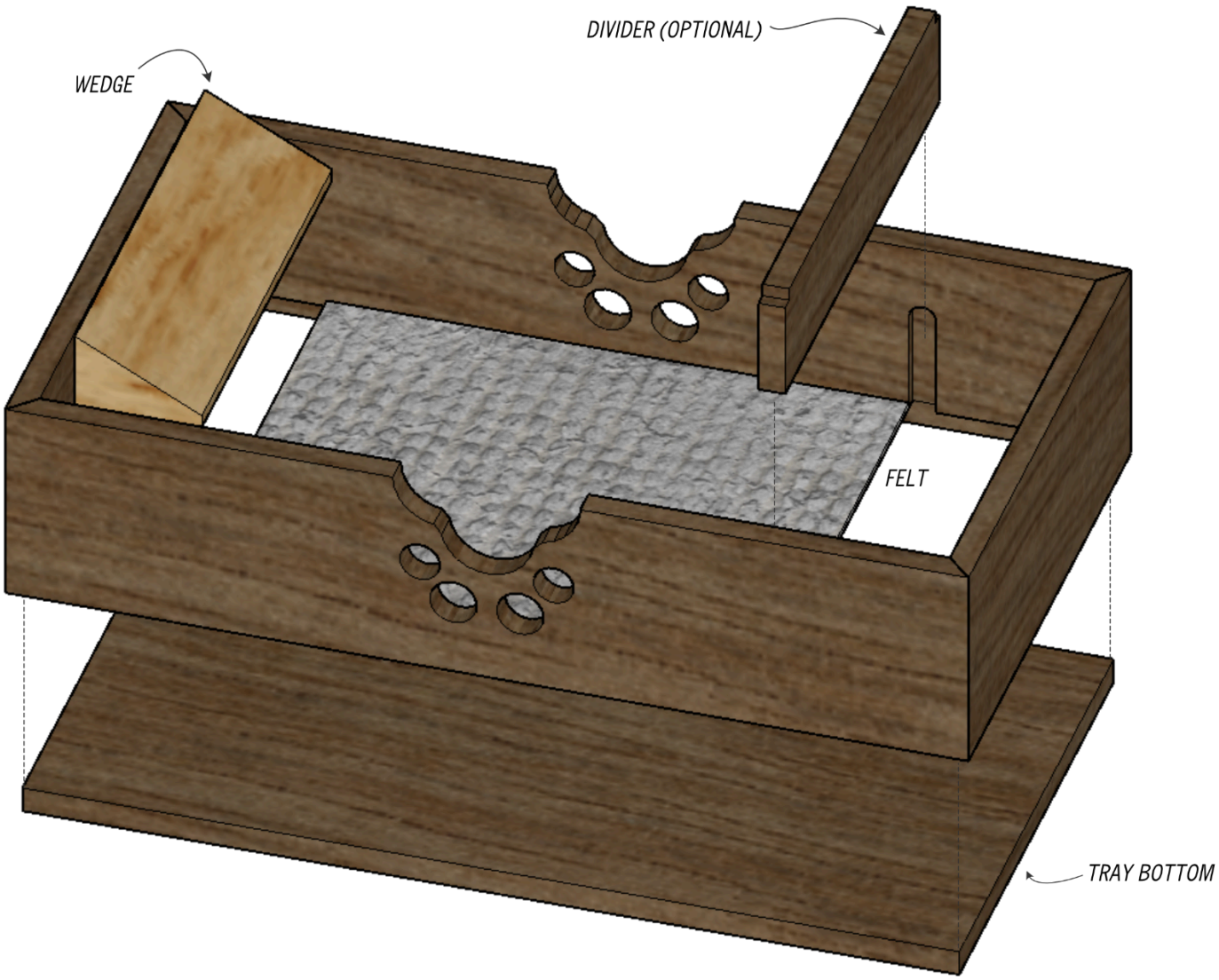


TRAY BUILD

CUT MITERS. Set up the table saw blade to 45°. Using a crosscut sled, cut miters on both ends of the long and short sides while also cutting them down to their final sizes:

- Long sides — 9”
- Short sides — 5”

DECIDE ORIENTATION. Choose the inside and outside faces, then mark the top and bottom edges. Match the grain across adjacent sides if possible. Label the orientation with pieces of masking tape.



CUT RABBET. Set up the router table with a ¼" bit. Since the parts are only ¼" thick, set the rabbet depth to ⅛". Cut ¼" x ⅛" rabbets on the inside bottoms of all four sides. Rabbets should run the full length of the sides.

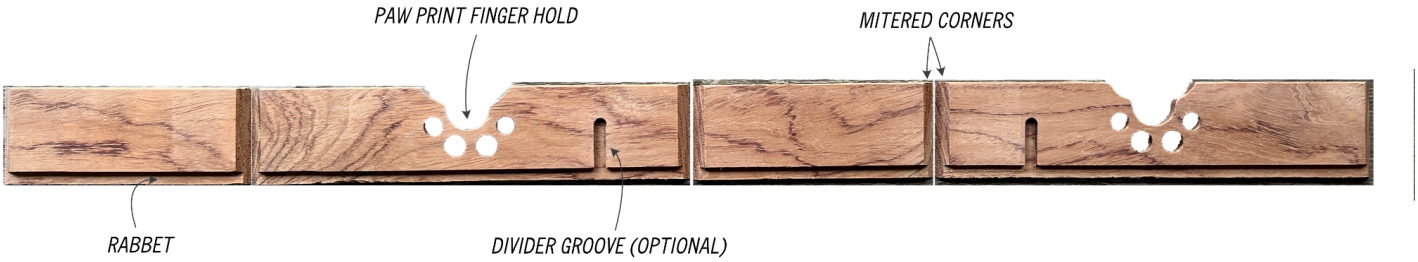


Photo: Inside faces of the tray sides before assembly

DIVIDER GROOVE (OPTIONAL). If you'd like to include a dice storage compartment inside the tray, cut the divider's groove at this time. Position the compartment towards the 'front' of the tray (relative to where the wedge will be installed later at the 'back' of the tray). The compartment should be 1 ½" wide to allow adequate space for dice storage. Thus, on each long side, measure 1 ½" from the inside point of the miter to mark the edge of the

divider groove. Using the same ¼" router bit, cut a ¼” wide groove that extends 1” upward from the rabbet (this is a tricky setup on the router table; a stop block can help halt the cut at exactly 1”). It’s okay to leave the top of the groove rounded since it will be hidden by the divider, which is 1/16" taller.

FINGER HOLDS/PAW PRINTS. Next, cut openings along the top edges of the longer sides so someone can grip and lift out the randomizer. The version shown uses a paw-print pattern; a simple semicircular notch would serve the same purpose.

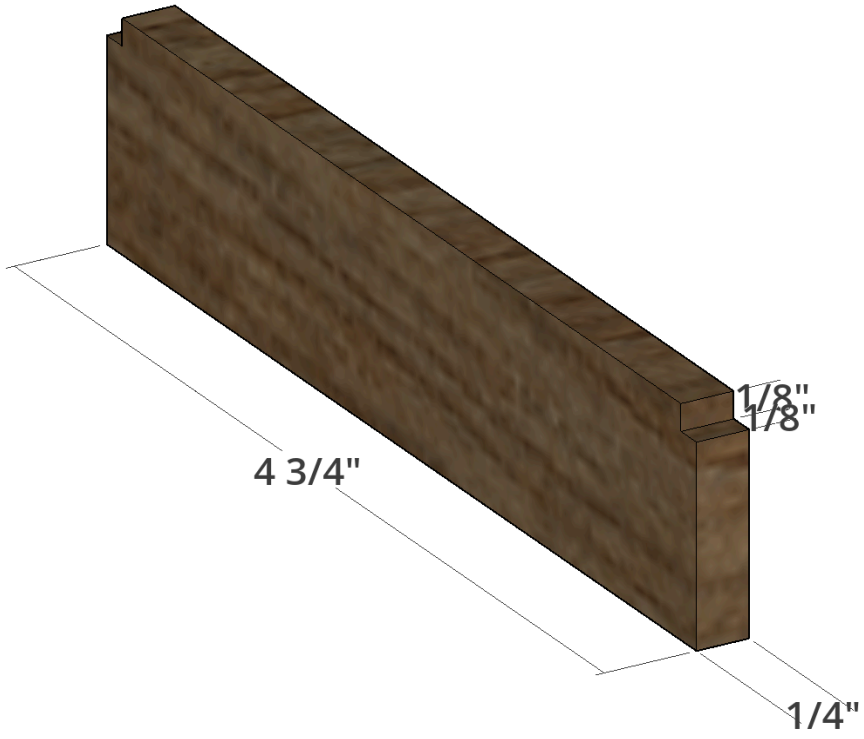
The paw print is made up of several circles of varying diameters:

- 3 paw circles — 7/8”
- 2 middle toes — 7/16”
- 2 little toes — 3/8”

Create a template by drawing the pattern on a scrap board. Exact spacing is not critical; arrange the circles to form a paw shape similar to the photo. Drill the scrap board to form a template. Use the 7/8” Forstner bit for the 3 large paw circles. Stack the two long sides and the scrap-board template while centering the paw-print along the length and secure everything with double-sided tape. Position the toe holes high enough that their lower edges remain above the top surface of the felt-liner after the bottom is installed. Transfer the pattern onto the actual tray sides by using the template's holes to guide the drill bit. Shape the paw with a spindle sander.

WARNING: *The drill bit can wander when drilling the two outside paw circles. Place a scrap piece up against the sides to give the drill bit some wood to bite into.*

DIVIDER. At the top of each end of the divider, notch away 1/8" so the lower portion fits into the ¼"-wide groove while the extra 1/16" of divider height covers the rounded end of the routed groove.



SIZING THE TRAY BOTTOM. Temporarily assemble the box using masking tape to hold the mitered sides together. This will help you size the bottom for an exact fit. Taking the rough-sized tray bottom from earlier, shave off small amounts with a table saw to sneak up on a perfect fit. The bottom should fit well, but not so tight as to force apart the mitered corners. Because the exact opening depends on the rabbets and miter fit, treat the cut-list dimension as oversize and fit the bottom to the dry-assembled tray.

SANDING INSIDE SURFACES. Sand interior faces prior to glue-up with 120-grit, followed by 220-grit. Sand the insides of the toe holes by wrapping a small piece of sandpaper around a pencil. Wipe away dust with a tack cloth or vacuum.

TRAY ASSEMBLY.

1. Dry-fit the tray and prepare the masking tape and/or band clamp.
2. Apply the first thin coat of glue to the miters; wait 1–2 minutes.
3. Apply the second coat to the miters, rabbets, and divider grooves.
4. Fold the sides closed; secure with masking tape.
5. Insert the divider, if used. Tap it in, so that the divider is seated flush.
6. Seat the bottom in the rabbets.
7. If using a band clamp, apply it now; then place a heavy weight on the bottom.
8. Allow at least one hour before removing clamps.

The first glue application seals the end grain so the second coat bonds properly. The masking tape should be enough to hold the mitered sides together, but a band clamp provides evenly distributed pressure across all four corners.

SPLINES. Once the glue has dried, cut splines to reinforce the mitered corners. Three splines per corner add strength and a decorative touch. Evenly space out the splines. Locate the upper and lower slots ⅜" from the top and bottom edges. Center the third slot halfway between them, 1" from either edge (see graphic).

Using a spline jig, cut the spline slots no more than ⅜” deep on each face. Deeper slots can break through into the tray interior and require trimming. Slots will be exactly ⅛" wide due to the kerf of the saw blade. Tape over the corners with masking tape to help reduce tearout.

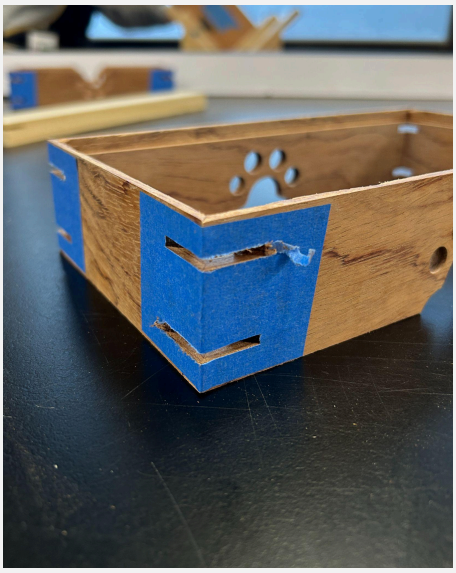
> ***NOTE:*** *The shown version had ¾” deep splines that protruded into the inside corners. Although aesthetically pleasing, it was cumbersome to trim them with a sharp 45° skew chisel afterwards.*

Rip ⅛"-thick strips of wood for the splines. Use a contrasting wood (matching randomizer back if possible). You might find it helpful to use a *thin-rip guide* and a *zero-clearance table saw insert* here.

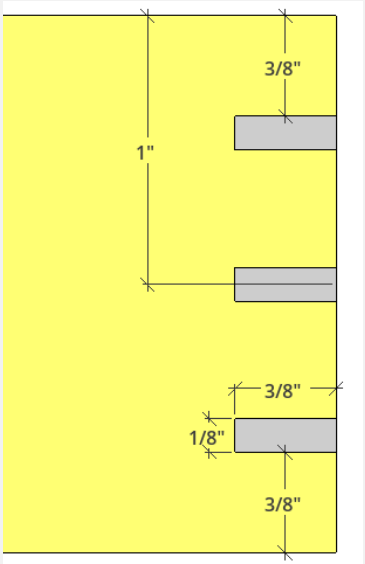
Cut the strips into oversized triangles using a bandsaw. Apply a thin layer of glue to each triangle and insert them snugly into the spline slots. Let the glue dry for 24 hours, then saw off the excess with a flush cut saw.



Use a spline jig for the table saw.



Tape the corners to reduce tearout.



Recommended spline spacing and measurements.

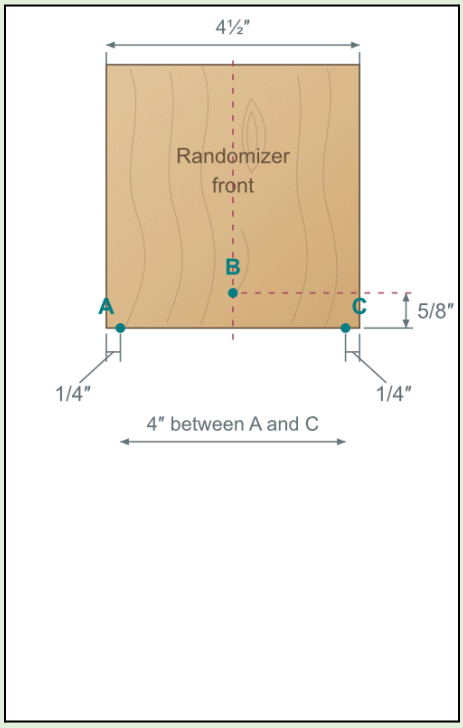
RANDOMIZER BUILD

CUT ARCHES ON THE FRONT. Cut two gentle arches on the randomizer front. The top arch is to make it easy to toss dice into the randomizer. The bottom arch frames the opening where the dice exit into the tray.

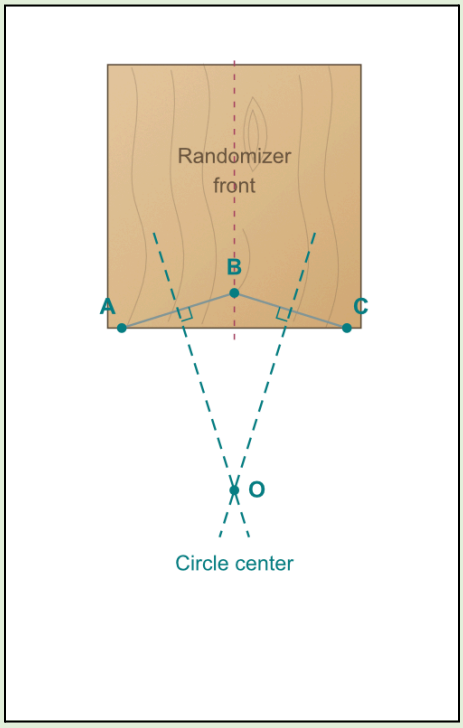
Mark ¼” from the edges and ⅝” from the top and bottom. Lay out each arch using the three-point construction shown below. Construct one arc at the top edge and one at the bottom edge. Cut with a bandsaw and smooth with a spindle sander.

How-To: Draw an Arch Through Three Points

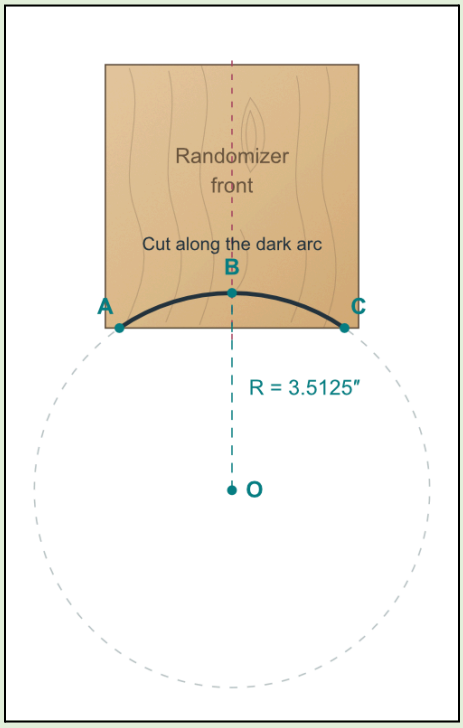
Bottom arch shown. Repeat the construction for the top arch of the randomizer front.



1. Mark three points. *Points A and C are ¼" from each side. Point B is centered, ⅝" above the bottom.*



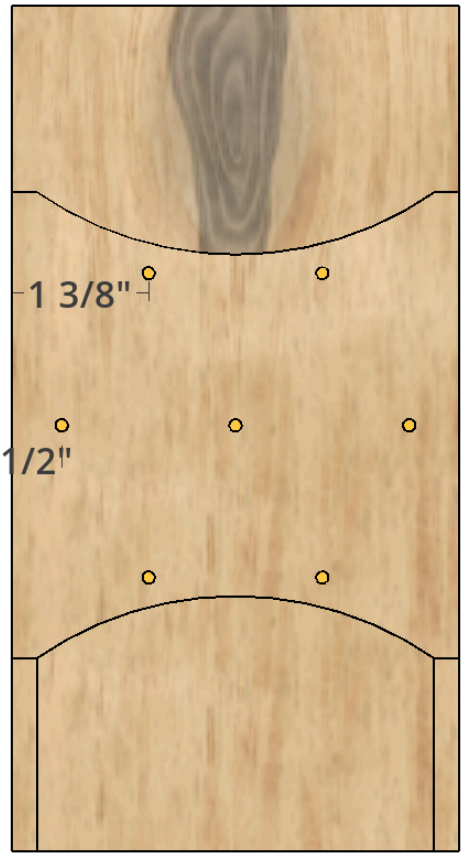
2. Find the circle's center. *Join A-B and B-C. Draw a perpendicular line through each segment's midpoint. Point O marks the circle's center.*



3. Draw the arch. *Place the compass point at O and draw an arc through A, B, and C. The circle center is 2.8875" below the bottom edge.*

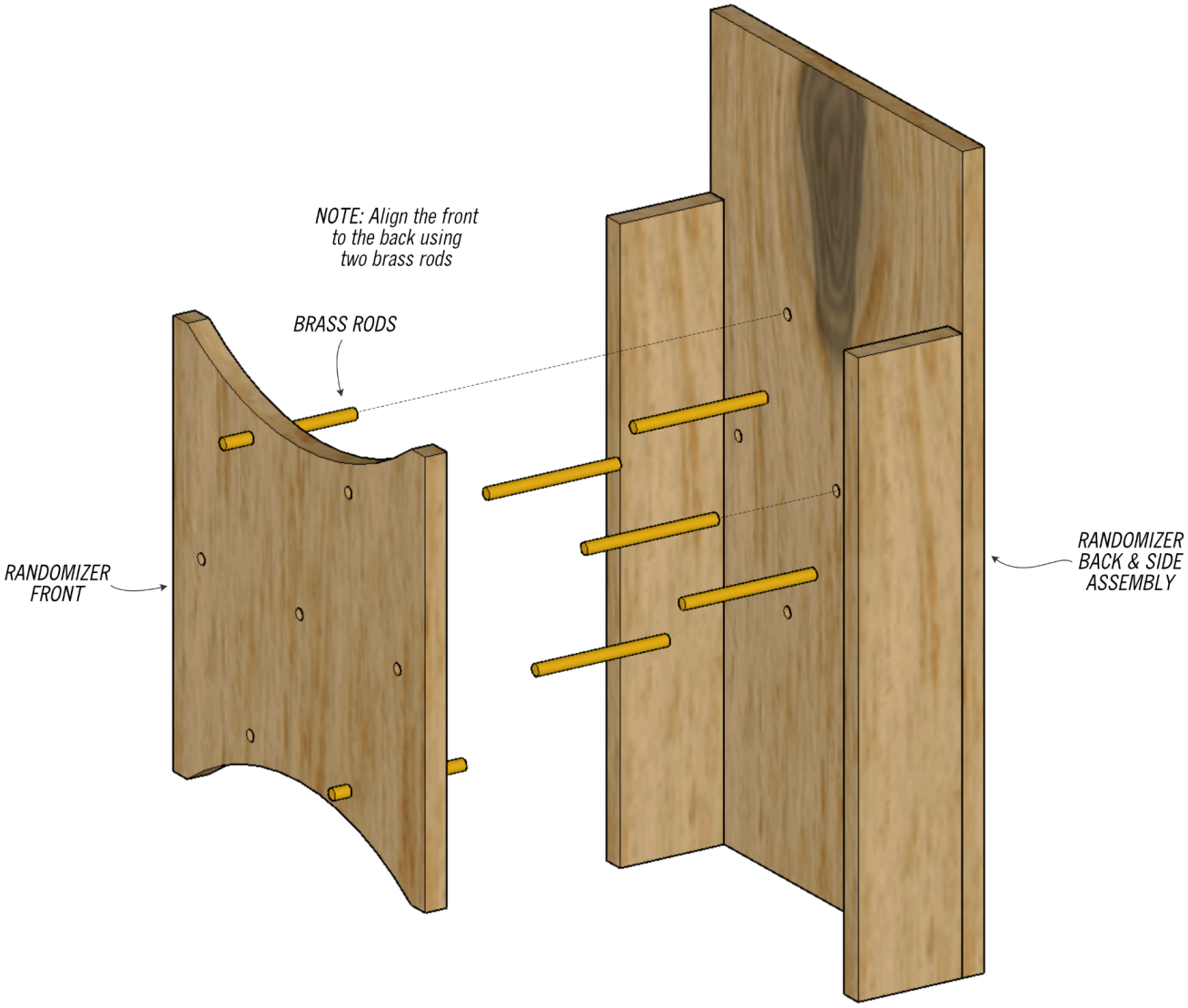
BRASS ROD HOLES. The brass rods give the dice something to hit and tumble off of as they fall through the randomizer. ⅛" brass rods can be purchased on Amazon.

Mark with a pencil seven points in a 2-3-2 pattern as shown in the diagram. Center the randomizer front on the randomizer back and double-sided tape the two pieces together. Using a drill press and ⅛" drill bit, drill the seven holes. Drill completely through the randomizer front and approximately $\frac{1}{16}$" into the randomizer back.



CUTTING BRASS RODS. From ⅛"-diameter brass rod stock, cut seven pieces approximately 1½" long. The exact length is not critical, but the rods should be long enough to span from the back holes through the front face. Clean up the rough edges left by the bolt cutters with 120-grit sandpaper. Chamfer one end of each rod with sandpaper to help the rods slip easily into the holes.

SANDING & FINISHING INSIDE SURFACES. Since the inside surfaces of the randomizer will be unreachable once it's glued, sand and finish them with Heritage oil now. **Do not** apply oil to the mating edges that will later be glued.



RANDOMIZER ASSEMBLY. Start by gluing the two randomizer sides to the randomizer back, forming a three-sided assembly. Leave the randomizer front off until you are ready to align it and install the brass rods.

Use CA glue (medium strength) and spray activator as well as standard wood glue to speed up the set time. Unlike the tray, the randomizer will be assembled with butt joints (the sides sit on top of the back). Gluing up a three-sided assembly can be tricky since there's nothing to hold the pieces at a perfect 90° angle. To solve this, use a flat vertical surface as a reference.

1. Lay the randomizer back flat on the table. Take one of the side pieces.
2. Apply two small drops of CA glue along one edge of the randomizer back, and a thin line of wood glue between them.
3. Press the randomizer side against the vertical reference to hold it square to the back, and hold steady for 10 seconds.
4. Repeat steps 2 and 3 for the other edge and second randomizer side.

Now you have a three-sided assembly.

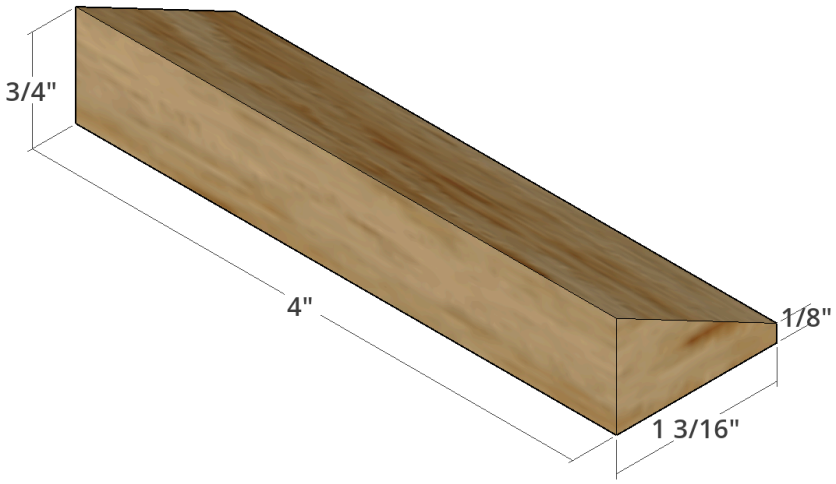
Next, glue the randomizer front. This step requires E6000 industrial-strength adhesive, which bonds metal to wood. The butt joints will be secured with both CA glue and wood glue. The CA glue acts as an instant clamp while the wood glue cures.

You have a 10-minute working window before the E6000 sets, so work quickly.

1. Again lay the three-sided assembly flat on the table, with the randomizer back lying down and the sides pointing up.
2. Drop blobs of E6000 into the seven holes of the randomizer back.
3. Apply two small drops of CA glue onto the long edges of each randomizer side, and a thin line of wood glue between them.
4. Spray activator on the randomizer front.
5. Insert two rods through the front, and lower the randomizer front slowly while aligning the rods into matching back holes (the CA glue sets in 1–2 seconds, so make sure everything is aligned).
6. Press firmly and hold for 10 seconds.
7. Push the remaining brass rods through the holes into the E6000.

The back ends of the rods are permanently bonded with E6000. The front holes act as snug locating sockets, so adhesive is not needed there. The randomizer body is now fully assembled.

SANDING OUTSIDE SURFACES. Sand the outside surfaces of the tray and the randomizer until smooth (an edge sander works well for this). Knock down all sharp edges.



CUT THE WEDGE. The wedge sits inside the tray at the base of the randomizer. Its 25° angle deflects the dice as they exit the randomizer, shooting them forward into the front of the tray. Without it, dice would fall straight down and sit trapped underneath the randomizer.

Start with a straight ¾"-thick blank at least 4½" long and wide enough to support safely against the fence. Place the fence on the left side of the blade, then stand the blank on its narrow edge (Step 1). Set the table saw blade angle to 25° so that it points toward the fence (Step 2). Cut the wedge by making several passes, raising the blade incrementally. Stop the angled passes while a thin web—approximately ⅛"—still connects the wedge to the blank. Do not cut the wedge free during the angled passes (Step 3).

CAUTION: *never raise the blade into the fence; use an auxiliary fence if needed. Keep a push pad between your fingers and the blade at all times.*

Move the fence to the right side of the blade, straighten the blade back to 0°, and cut off the wedge (Step 4).

Cut the wedge to its length so that it fits snugly between the randomizer walls—roughly 4”. Sand it smooth and knock down sharp edges.

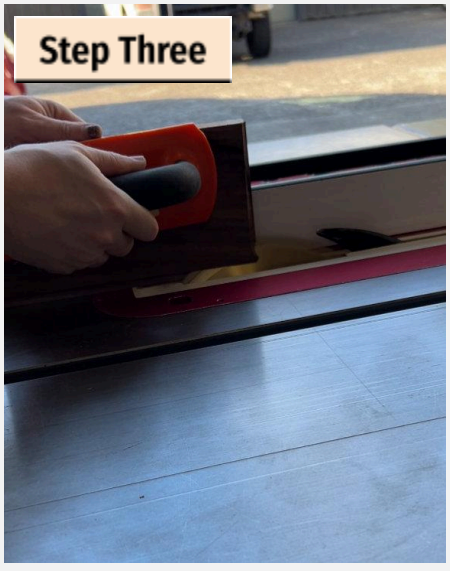
To attach the wedge, stand the randomizer upright on its end inside the tray to correctly position the wedge. Place the wedge between the randomizer walls and draw a line to mark its front edge. Remove the randomizer, then apply blobs of CA glue and wood glue inside the tray at the marked location. Spray activator on the bottom of the wedge and press it down firmly for 10 seconds to set.



Fence left of blade.



Tilt blade to 25°.



Make progressive angled cuts.



Return blade to 0° and cut wedge free.

FINISHING

Apply three coats of Heritage oil to all surfaces. Heritage is a penetrating natural oil finish; other penetrating oil finishes like tung oil or Danish oil work too. Be careful not to get any oil on the tray bottom interior. A little finish around the edges is fine, but the center needs to remain bare so the felt glue can adhere properly. Let the finish dry for 24 hours.

FELT. The felt lines only the main rolling area of the tray — not behind the wedge or inside the dice storage compartment. Measure from the front edge of the wedge to the divider (or to the opposite tray end if you skipped the divider). The felt should fit tightly to the tray sides. Cut the felt with sharp scissors or a razor blade. Spray heavy-duty Gorilla Glue adhesive on the backside of the felt and gently adhere it to the tray bottom.